

TU DORTMUND

CASE STUDY II: PROJECT 1

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Final Report: Styrian Health Survey Data

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Chapter 1

Introduction

This chapter introduces the Styrian Health Data case study, states the analytical objectives, and summarises the dataset and principal findings that are developed in subsequent chapters.

1.1 Aims and Goals

The project pursues three interconnected statistical objectives:

1. **Characterise data quality** and prepare an analysis-ready sample through systematic cleaning, missing-value handling, and plausibility checks.
2. **Identify determinants of blood pressure** using descriptive statistics, group comparisons, ordinary least squares (OLS) regression, and Random Forest models; evaluate whether self-reported wellbeing and clinical risk flags explain variation in systolic and diastolic blood pressure after adjustment for demographics.
3. **Extend the analysis spatially and contextually** by linking individual records to regional gross domestic product (GDP), urban–rural typology, and neighbourhood structure; test for neighbourhood effects on cardiovascular indicators.

Beyond these substantive goals, the project aims to compare regression and classification formulations, quantify predictive limits of available covariates, and document limitations inherent in voluntary kiosk data.

1.2 Data Description

This section describes the data collection process, the variable structure, and the data quality issues.

1.2.1 Collection

The data were **not** generated by a designed experiment. They arise from an **observational, voluntary health-screening programme** at public exhibition terminals [1]. Participants self-reported demographic and health information via interactive terminals; blood pressure was measured automatically by the kiosk equipment. Measurements were recorded at **three terminals** between **29 April 2006** and **29 October 2006**. The total number of observations in the dataset is **16,386** records with 19 attributes.

1.2.2 Variable Structure

Table 1.1 summarises the variable structure. Measured systolic and diastolic blood pressure (messwert_bp_sys/dia) serve as the primary outcome variables for most analyses, with self-estimated blood pressure values (schaetzwert_bp_sys/dia) available as auxiliary measures.

Independent (predictor) variables include demographics, self-reported health and risk factors, geography, and metadata such as timestamp and terminal number. External covariates merged in later stages include hourly temperature and humidity from the Open-Meteo Historical Weather API [4], regional GDP per capita from Statistics Austria [5], and urban–rural typology.

Table 1.1: Overview of key variables in the Styrian Health Data (2006).

Category	Variables and Description
Identifiers	Unique record ID (id), timestamp of entry (zeit), and data collection terminal (terminal)
Geography	Place of residence and administrative hierarchy: postal code, municipality (gemeinde), district (bezirk), and state (bundesland)
Health Status	Self-reported health condition (befinden), measured on an ordinal scale from 1 (very good) to 5 (very poor)
Demographics	Year of birth (geburtsjahr), gender (geschlecht), and derived age variable
Risk Factors	Smoking status (raucher), known diabetes status (blutzucker_bekannt), known cholesterol status (cholesterin_bekannt), and ongoing treatment status (in_behandlung)
Blood Pressure	Estimated systolic and diastolic blood pressure (schaetzwert_bp_sys/dia), and measured systolic and diastolic blood pressure (messwert_bp_sys/dia)

1.2.3 Data Quality Issues

Several quality issues were identified during exploratory analysis and cleaning:

- **Missing values:** Column-wise null counts include 331 missing state codes, 45–56 missing estimated blood pressure values, and 23 records with joint missingness in befinden, birth year, gender, and age.
- **Geographic representativeness:** Approximately 89% of records with state information originate from Steiermark; the sample is not nationally representative.

1.3 Main Results at a Glance

- **Age** is the dominant blood pressure predictor (Random Forest importance ≈ 0.67 ; OLS $\beta = +0.32$ mmHg systolic per year). Male sex, known blood sugar, and under treatment are associated with higher adjusted blood pressure; self-reported wellbeing is not significant after covariate adjustment.
- Individual-level blood pressure prediction remains weak (systolic $R^2 = 0.12$ – 0.19 ; diastolic $R^2 = 0.02$ – 0.10), with mean absolute errors of 10–14 mmHg.

- Regional GDP does not significantly predict state-level mean blood pressure. GDP is strongly negatively associated with overweight prevalence at NUTS level 2 ($r = -0.901$). Neighbourhood mean systolic pressure enters neighbourhood effect averaging problem (NEAP) models with a positive coefficient, but the intraclass correlation coefficient (ICC, 1.8%) indicates that most variance remains at the individual level.

1.4 Report Structure

The remainder of this report is organised as follows:

- Chapter 2 presents exploratory data analysis (Section 2).
- Chapter 3 describes the Random Forest modelling pipeline, performance comparisons, and classification extension.
- Chapter 4 reports regional and neighbourhood-level spatial analysis.
- Chapter 5 summarises the principal findings, limitations, and implications.
- Appendix B contains the artificial intelligence usage declaration.

Detailed statistical results and interpretation follow in the subsequent chapters; this introduction establishes context, data characteristics, and the analytical framework only.

Chapter 2

Data Analysis

This chapter presents exploratory data analysis of the Styrian Health Data at the individual level, covering data preparation, descriptive patterns, and group comparisons that inform the modelling in Chapter 3.

2.1 Data Preparation and Quality Assessment

The raw dataset contains 16,386 records with heterogeneous missing values. After removing incomplete and implausible entries and restricting ages to the range 13–100 years, a final analysis sample of 15,219 observations remains (7.1% excluded).

Key data quality characteristics are summarised below:

- 23 records are missing all core demographic and health variables (`befinden`, birth year, gender, and age).
- Column-wise missingness includes 331 missing state identifiers and 45–56 missing blood pressure estimates.
- Temporal coverage spans April to October 2006, with 373 measurements recorded before 10:00 and 116 after 18:00.
- Geographic coverage is concentrated in Styria, accounting for approximately 89% of records with valid state information.

Mean imputation for missing values was considered but not applied, as it was deemed likely to introduce bias into subsequent analyses.

2.2 Demographics and Health Status

Self-reported health status (`befinden`, scale 1–5) is heavily concentrated in the lower categories, as shown in Table 2.1.

The correlation structure of the numerical variables indicates generally weak to moderate linear relationships, visualised in Figure 2.1. The strongest correlation is observed between systolic and diastolic measured blood pressure ($r \approx 0.67$), while estimated and measured blood pressure values also show moderate association. Age exhibits only weak correlations with other variables, and health status shows minimal linear association with the remaining features.

Missing values were not fully random. Records with missing self-reported health condition often had missing values in other fields as well, indicating incomplete participant records rather than

Table 2.1: Distribution of self-reported health status (befinden).

Health Level	Count
1	5,897
2	7,931
3	2,240
4	198
5	97

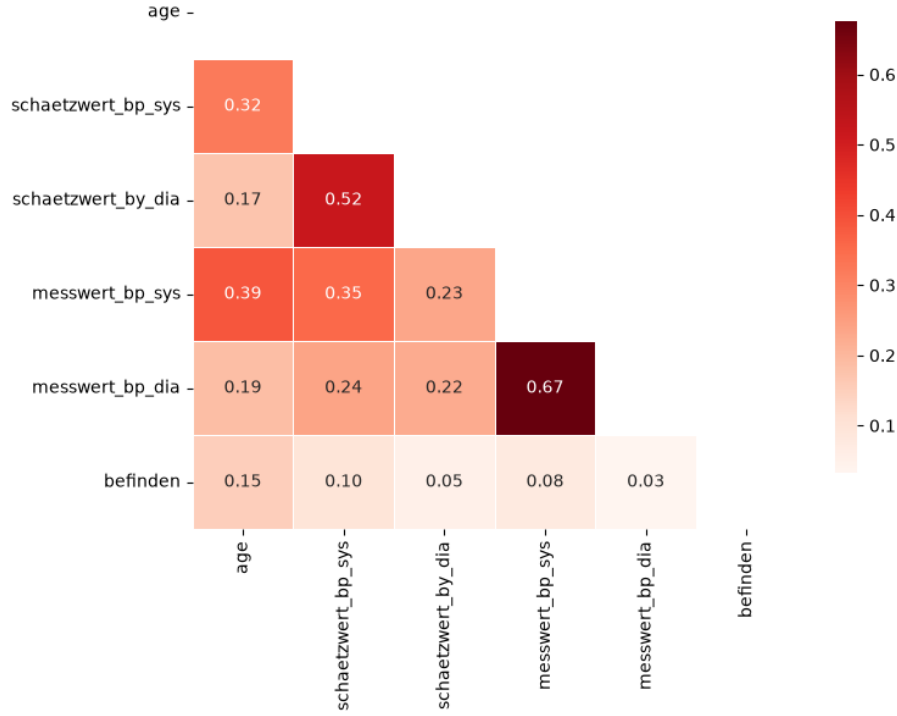


Figure 2.1: Lower-triangular correlation heatmap of selected numerical variables. Darker shading indicates stronger positive or negative Pearson correlation.

isolated recording errors. Most participants were between 30 and 70 years old; Figure 2.2 shows that the age distribution reflects a predominantly adult sample, which is appropriate for studying blood pressure variation.

The data also contain implausible entries. For example, there are 11 smokers below the age of 5 and birth years extending as far back as 1880, which likely indicate data entry errors rather than genuine observations.

2.3 Measured and Self-Estimated Blood Pressure

Figure 2.3 compares the distributional shapes of measured and self-estimated blood pressure. Density is defined such that the area under each curve equals one, allowing a comparison of shapes independent of sample size differences across groups.

The distribution of measured blood pressure is notably more dispersed than that of self-estimated values, particularly for systolic pressure. This suggests that participants tend to report values clustered around common or rounded reference points, whereas the measured data exhibit sub-

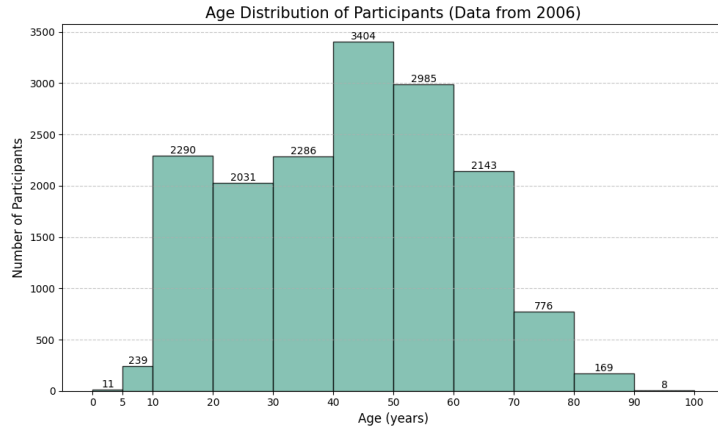


Figure 2.2: Age distribution of participants in the analysis sample ($n = 15,219$).

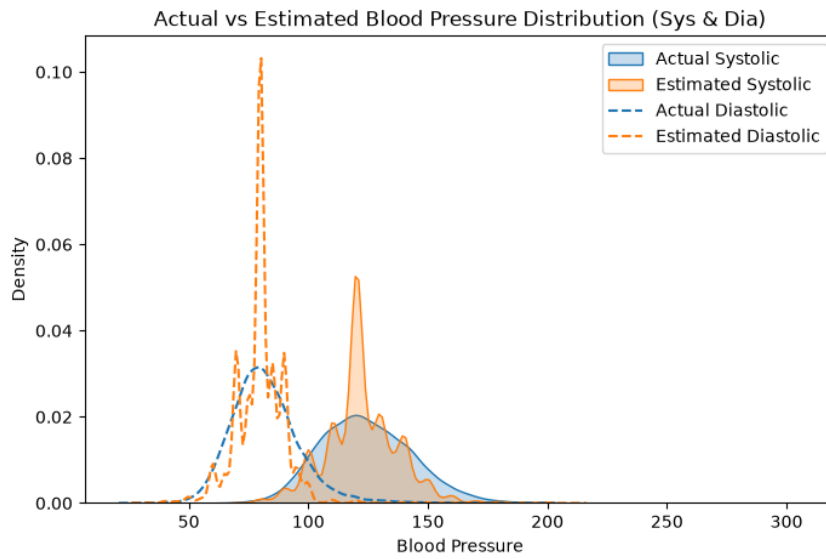


Figure 2.3: Kernel density estimates of measured and self-estimated systolic and diastolic blood pressure.

stantially greater variability.

2.4 Group Differences and Correlations

Figure 2.4 shows that male participants exhibit higher average systolic and diastolic blood pressure compared to female participants, although both groups display considerable internal variability.

The correlation analysis indicates a strong positive association between measured systolic and diastolic blood pressure. Age shows a moderate correlation with both outcomes. Self-estimated blood pressure values are also positively correlated with measured values; however, these relationships are weaker than the correlation observed between the two measured blood pressure variables.

2.5 Risk Factor Comparisons

Group comparisons confirm several statistically significant differences, summarised in Table 2.2.

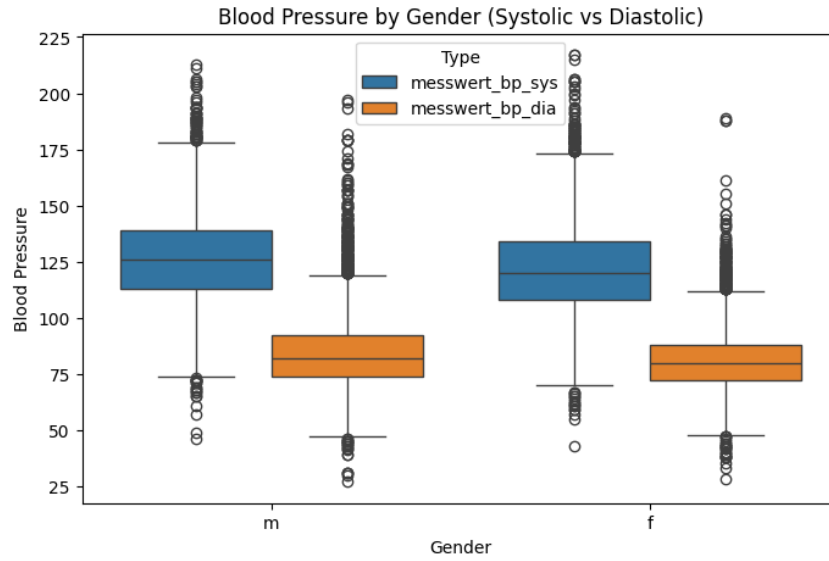


Figure 2.4: Blood pressure and self-reported health status by gender. Box plots summarise the central tendency and spread within each group.

Table 2.2: Mean blood pressure differences across risk groups. Values are group means in mmHg; p denotes the two-sample test significance level.

Comparison	Systolic	Diastolic
Male vs. Female	128.0 vs. 122.8 ($p < 0.001$)	84.6 vs. 81.0 ($p < 0.001$)
Smoker vs. Non-smoker	122.9 vs. 125.5 ($p < 0.001$)	82.5 vs. 82.6 (not significant)
Blood sugar known vs. not	128.8 vs. 123.7 ($p < 0.001$)	83.8 vs. 82.1 ($p < 0.001$)
Cholesterol known vs. not	128.0 vs. 123.4 ($p < 0.001$)	83.5 vs. 82.0 ($p < 0.001$)
In treatment vs. not	137.8 vs. 123.2 ($p < 0.001$)	86.4 vs. 82.0 ($p < 0.001$)

Higher blood pressure among individuals with known conditions likely reflects confounding due to increased medical awareness and pre-existing health conditions rather than a direct causal effect of the awareness indicator itself.

Chapter 3

Modelling

This chapter describes the predictive modelling of blood pressure using Random Forest methods. The analysis builds on the cleaned dataset described in Chapter 2 ($n = 15,219$) and progresses from a baseline multi-output regressor through feature engineering, robustness checks, and a classification extension. Implementation details are documented in `modeling.ipynb`.

3.1 Modelling Pipeline Overview

The pipeline comprises four stages:

1. **Data preparation** — encode categorical and boolean predictors as numeric features; define systolic and diastolic blood pressure as joint regression targets.
2. **Baseline regression** — train a multi-output Random Forest on demographic and clinical covariates only.
3. **Model refinement** — apply age-stratified splitting, cross-validation, tree-depth tuning, and augment features with temporal and weather variables.
4. **Evaluation and extensions** — compare models on held-out test data; add hypertension classification and residual diagnostics.

The primary modelling goal is to predict measured blood pressure from available covariates. A secondary goal is to classify elevated blood pressure categories when exact mmHg prediction proves unreliable. Model performance is assessed using the coefficient of determination (R^2) and mean absolute error (MAE), defined for outcome y_i and prediction $\hat{y}_i$ as

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}, \quad \text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|, \quad (3.1)$$

where n denotes the number of test observations and $\bar{y}$ is the sample mean of the outcome.

3.2 Feature Set and Encoding

For simplicity, all model inputs are transformed to numeric form. Categorical and boolean variables (gender, smoker status, known blood sugar, known cholesterol, under treatment) are one-hot encoded with `drop_first=True`. The baseline predictor set comprises seven features: age, health condition (`befinden`), gender, smoker status, known blood sugar, known cholesterol, and under treatment. Two blood pressure outcomes are predicted jointly: systolic (`bp_systolic`) and diastolic (`bp_diastolic`).

3.3 Train–Test Splitting

The initial baseline used a standard 80/20 split. In the refined pipeline, splitting is **stratified by age group** (0–20, 21–40, 41–60, 61–80, 81–100 years) to preserve the age distribution across training and test sets (12,175 / 3,044 records). This addresses the concern that a random split could produce unrepresentative test cohorts in a dataset dominated by middle-aged participants.

3.4 Weather Data Integration

Hourly temperature and relative humidity for Graz, Austria (2006) are retrieved from the Open-Meteo Historical Weather API [4] and matched to each measurement timestamp rounded to the nearest hour. One hundred twenty-three records fall outside the weather coverage period and are excluded, leaving $n = 15,096$ for weather-enhanced models.

3.5 Baseline Model: Multi-Output Random Forest

The baseline model is a **multi-output Random Forest regressor** that predicts systolic and diastolic blood pressure simultaneously from the seven baseline features. Hyperparameters are held constant across iterations unless noted:

- `n_estimators = 500, max_features = 'sqrt', min_samples_leaf = 5, random_state = 42.`
- Baseline `max_depth = None` (unlimited tree depth).

On the stratified test set, the baseline achieves weak explanatory power (Table 3.2). Systolic predictions show a modest positive trend with actual values; diastolic predictions are substantially noisier. Mean absolute errors of 10–14 mmHg are clinically meaningful: a typical prediction may deviate from the true value by an amount relevant to hypertension diagnosis.

Feature importance confirms that **age dominates** (importance ≈ 0.68), followed by under treatment (≈ 0.11), gender (≈ 0.08), and health condition (≈ 0.06). Smoker status, blood sugar awareness, and cholesterol awareness contribute least (< 0.03 each). Lifestyle and clinical-awareness variables therefore add limited predictive information relative to age.

As a supplementary check, **single-output models** that include the complementary blood pressure component as a predictor perform substantially better ($R^2 \approx 0.48$ – 0.53). This reflects the strong correlation between systolic and diastolic readings ($r \approx 0.66$) rather than improved prediction from demographic covariates alone. Single-output formulations are not pursued further because they are not applicable when both pressures must be predicted from non-blood-pressure features.

3.6 Cross-Validation and Overfitting Control

Five-fold cross-validation on the full feature set yields mean $R^2 = 0.140$ (± 0.013) for systolic and 0.035 (± 0.014) for diastolic blood pressure, confirming that baseline performance is stable across folds rather than an artefact of a single split.

Tree depth is tuned via cross-validation on systolic R^2 ; results are summarised in Table 3.1.

Moderate depth (`max_depth = 10`) performs best. Unlimited or very deep trees (`None`, `20`) match shallower cross-validated performance but show signs of overfitting relative to the depth-10 setting. All subsequent refined models use `max_depth = 10`.

Table 3.1: Cross-validated systolic R^2 by maximum tree depth (`modeling.ipynb`).

Maximum depth	Mean CV R^2
3	0.138
5	0.157
10	0.163
20	0.140
None (unlimited)	0.140

3.7 Temporal Feature Engineering

Four temporal variables are extracted from the measurement timestamp: hour of day, day of week, month, and season (one-hot encoded). Combined with the baseline covariates, the **temporal Random Forest** improves test-set performance, with the largest gain for diastolic blood pressure (Table 3.2).

In the temporal model, age remains the strongest predictor (importance ≈ 0.42), but hour (≈ 0.09), day of week (≈ 0.08), and month (≈ 0.06) contribute non-trivial information. Permutation importance confirms age as the dominant variable while temporal features provide incremental value.

3.8 Weather Feature Engineering

Temperature (2 m) and relative humidity (2 m) are added to the temporal feature set. The **weather-enhanced Random Forest** yields the best regression performance overall, as shown in Table 3.2.

Table 3.2: Regression model comparison on held-out test sets (`modeling.ipynb`).

Model	Sys. R^2	Sys. MAE	Dia. R^2	Dia. MAE
Baseline multi-output RF	0.130	14.14	0.042	10.50
Temporal RF	0.176	13.81	0.086	10.19
Weather-enhanced RF	0.192	13.60	0.097	10.05

Weather integration reduces mean absolute error by approximately 0.5 mmHg (systolic) and 0.4 mmHg (diastolic) relative to the baseline. The improvement is consistent but modest: even the best model explains only 19% of systolic variance and 10% of diastolic variance. Figure 3.1 shows that scatter plots of actual versus predicted values remain widely dispersed.

Figure 3.2 confirms that age remains the dominant predictor in the weather-enhanced specification; temporal and weather variables provide supplementary information but do not overturn the age gradient.

Scenario simulations with the weather-enhanced model hold all covariates fixed except one variable. Predicted blood pressure shows a weak inverse association with temperature and modest within-day variation by hour. These patterns are **predictive associations**, not evidence of causal effects.

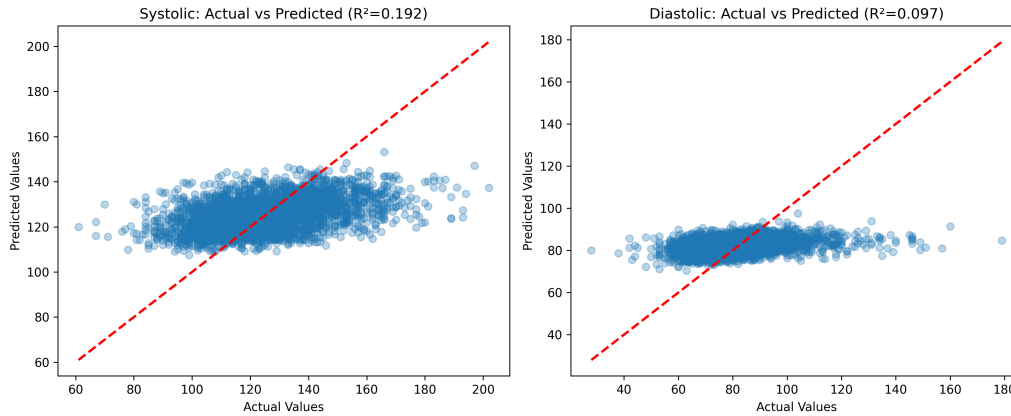


Figure 3.1: Actual versus predicted blood pressure for the weather-enhanced Random Forest on the held-out test set. Dashed lines indicate perfect prediction.

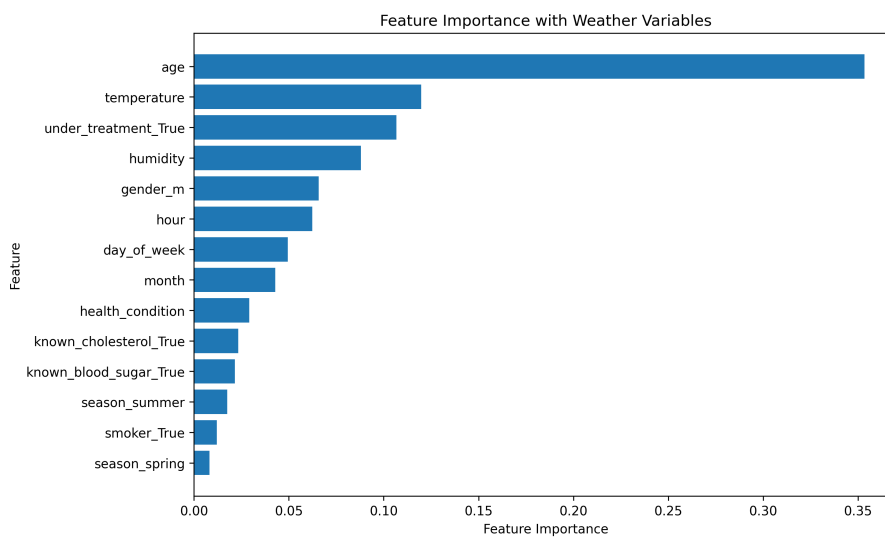


Figure 3.2: Permutation-based feature importance in the weather-enhanced Random Forest (`modeling.ipynb`).

3.9 Classification Extension

Elevated blood pressure is defined as systolic ≥ 140 mmHg or diastolic ≥ 90 mmHg. Separate binary targets are constructed for systolic and diastolic elevation. A weighted Random Forest classifier (`class_weight = 'balanced'`) addresses class imbalance. Features match the weather-enhanced regression model; splitting is stratified on the classification target. Performance on held-out test data is summarised in Table 3.3.

Table 3.3: Hypertension classification performance on held-out test data (`modeling.ipynb`).

Target	Accuracy	Precision	Recall	F1
Systolic (≥ 140 mmHg)	0.712	0.394	0.560	0.463
Diastolic (≥ 90 mmHg)	0.640	0.367	0.519	0.430

Systolic classification achieves higher accuracy and F1, but both targets exhibit **low precision** (39% and 37%). The classifiers detect a reasonable share of elevated cases (recall 52–56%) at the cost of many false positives. Figure 3.3 shows that lowering the classification threshold from 0.5

to 0.4 increases recall and F1 slightly, while further reduction trades precision for additional false alarms. The choice of threshold therefore depends on whether false negatives or false positives carry greater cost in the intended application.

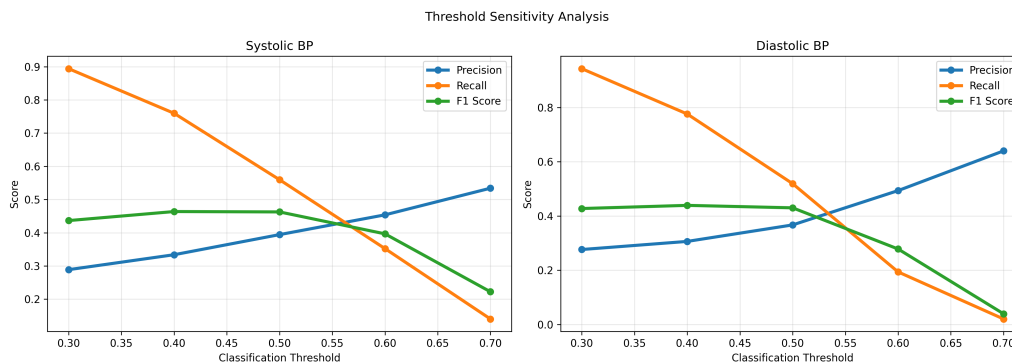


Figure 3.3: Precision, recall, and F1 score across classification thresholds for systolic and diastolic hypertension targets (`modeling.ipynb`).

3.10 Residual Diagnostics

Residuals from the weather-enhanced regression model are examined to identify systematic prediction failures. Residuals are approximately centred across temperature values, indicating no strong temperature-related bias. Systolic residuals exhibit larger variance than diastolic residuals, particularly for middle-aged and older participants, suggesting that age-related heterogeneity remains even after modelling.

3.11 Summary and Interpretation

The modelling analysis yields three principal conclusions:

- **Limited predictive power.** Available covariates explain a small fraction of blood pressure variance. The best regression model achieves $R^2 = 0.19$ (systolic) and 0.10 (diastolic), with mean errors of 10–14 mmHg. Models capture population-level trends—especially the age–blood pressure relationship—more reliably than individual readings.
- **Age dominates; refinements add modest value.** Age accounts for most feature importance in every model specification. Temporal and weather features improve performance consistently but do not overturn the dominance of age. Tree-depth tuning and stratified splitting improve methodological rigour without fundamentally changing this conclusion.
- **Classification offers a complementary perspective.** Hypertension classifiers achieve moderate accuracy (64–71%) but low precision, reflecting both class imbalance and limited feature informativeness. They are better suited to risk stratification than to precise clinical diagnosis.

These results complement the inferential findings in Chapter 2. Random Forest and OLS approaches converge on age and treatment status as the strongest associations, while lifestyle variables contribute comparatively little to prediction.

Chapter 4

Spatial Analysis

This chapter links individual blood pressure records to regional socioeconomic indicators and neighbourhood context. It asks whether regional prosperity and health-style markers relate to average blood pressure, and whether neighbourhood mean pressure adds explanatory value beyond individual covariates.

4.1 Objectives

Three questions guide this extension:

1. Does regional GDP per capita relate to average blood pressure?
2. Do regional health-style indicators (overweight prevalence, average body mass index) relate to average blood pressure?
3. Does neighbourhood mean blood pressure influence individual readings after adjustment for individual risk factors (the neighbourhood effect averaging problem, NEAP)?

4.2 Data Sources and Geographic Coverage

Regional indicators are drawn from open sources: GDP per capita and disposable household income at NUTS level 2 from Statistics Austria [5], and state-level average body mass index together with overweight prevalence (share of population with body mass index ≥ 25) from the Austrian cardiovascular risk monitoring report [2]. Administrative boundaries for choropleth maps are taken from published Austria shapefiles [3].

For ecological comparisons, mean systolic and diastolic blood pressure are computed separately for each state (`bundesland`) and merged with these regional series. Nine states enter the analysis. As noted in Chapter 2, the underlying sample is strongly concentrated in Styria (13,605 of 15,219 records; 89%). Remaining observations are spread across eight other states with counts from 27 (Vorarlberg) to 476 (Wien). Nationwide Austrian population studies document substantial regional variation in blood pressure [6]; the state-level aggregates reported here are derived from the voluntary screening sample rather than from those representative surveys.

Two limitations follow directly from this design. First, with only nine spatial units, regional correlations have low statistical power. Second, associations at region level need not hold for individuals within regions (ecological fallacy). Results below are therefore interpreted as contextual patterns, not as individual-level causal effects.

4.3 Regional GDP, Income, and Blood Pressure

Pearson correlation between state-level GDP per capita and mean blood pressure is negative but not statistically significant:

- Systolic: $r = -0.433$, $p = 0.245$.
- Diastolic: $r = -0.325$, $p = 0.393$.

Choropleth maps (Appendix A, Figures A.1a–A.4) show clear regional variation in GDP, but no corresponding gradient in mean blood pressure across richer and poorer states. Disposable income was mapped descriptively at regional level; no formal correlation with blood pressure was computed. Given the GDP results and the small number of regions, a strong income–blood pressure association at state level is not supported by the available evidence.

4.4 Health-Style Indicators at Regional Level

State-level average body mass index shows a weak, non-significant positive correlation with mean diastolic blood pressure ($r = 0.222$, $p = 0.567$). By contrast, GDP per capita and overweight prevalence are strongly negatively correlated ($r = -0.901$, $p = 0.001$): states with higher GDP report lower overweight prevalence. Overweight prevalence was not significantly associated with mean blood pressure in this analysis. The significant GDP–overweight link nevertheless indicates that regional economic context relates to population-level health-style markers, even though GDP does not directly predict regional mean blood pressure here.

4.5 Neighbourhood Effects

The neighbourhood effect averaging problem (NEAP) arises when group means are used as predictors alongside individual-level variables. Neighbourhoods are defined by postal code (`postleitzahl`). Codes with ten or fewer records are excluded so that neighbourhood means are based on sufficient data. This filter retains $n = 13,632$ individuals.

For each retained record, neighbourhood mean systolic blood pressure $\bar{y}_{g(i)}$ is the average `messwert_bp_sys` among all individuals sharing postal code $g(i)$. Two OLS models regress individual diastolic blood pressure y_i on individual covariates:

$$y_i = \beta_0 + \beta_1 \text{age}_i + \beta_2 \text{smoker}_i + \beta_3 \text{chol}_i + \varepsilon_i, \quad (4.1)$$

$$y_i = \beta'_0 + \beta'_1 \text{age}_i + \beta'_2 \text{smoker}_i + \beta'_3 \text{chol}_i + \beta'_4 \bar{y}_{g(i)} + \varepsilon'_i, \quad (4.2)$$

where age_i is age in years, smoker_i and chol_i are indicator variables for smoker status and cholesterol awareness, $\bar{y}_{g(i)}$ is the neighbourhood mean systolic blood pressure, and ε_i , ε'_i are error terms. Equation (4.1) is Model 1 (without neighbourhood); Equation (4.2) is Model 2 (with neighbourhood).

Model 1 explains little variance ($R^2 = 0.021$). Age ($\beta_1 = 0.117$, $p < 0.001$) and smoker status ($\beta_2 = 0.904$, $p = 0.007$) are positively associated with diastolic pressure; cholesterol awareness is not significant ($p = 0.217$).

Model 2 adds neighbourhood mean systolic blood pressure with $\beta'_4 = 0.383$. The intercept falls from 77.15 to 29.47, indicating that much of the baseline diastolic level in Model 1 is absorbed by the neighbourhood term. Age and smoker status remain positively associated with diastolic pressure. Coefficient estimates for both models appear in Table 4.1.

Table 4.1: NEAP regression coefficients for diastolic blood pressure (Equations (4.1)–(4.2); `spatial_analysis_part1.ipynb`).

Predictor	Model 1	Model 2
Intercept	77.15	29.47
Age (per year)	0.117	0.109
Smoker	0.904	1.000
Cholesterol awareness	0.328	0.392
Neighbourhood mean systolic BP	—	0.383

To measure clustering across neighbourhoods, a linear mixed model with a random intercept $u_g \sim N(0, \sigma_u^2)$ by postal code is fitted for systolic blood pressure. The intraclass correlation coefficient is

$$\text{ICC} = \frac{\sigma_u^2}{\sigma_u^2 + \sigma_\varepsilon^2} = 0.018, \quad (4.3)$$

where σ_u^2 is the between-neighbourhood variance and σ_ε^2 is the within-neighbourhood residual variance. Roughly 1.8% of systolic blood pressure variance lies between postal codes and 98.2% within codes. Neighbourhood mean systolic pressure therefore enters Model 2 with a positive coefficient, but postal-code membership accounts for only a small share of total variance.

4.6 Summary

Regional GDP does not significantly predict state-level mean blood pressure. At regional level, economic prosperity is strongly linked to lower overweight prevalence, but not to lower mean blood pressure. NEAP analysis shows a modest contextual association between neighbourhood mean systolic pressure and individual diastolic pressure, while the low ICC in Equation (4.3) confirms that individual-level factors dominate. Inferential conclusions about blood pressure determinants should rely on the individual-level results in Chapter 2; the spatial findings provide geographic context rather than primary evidence for prediction or causation.

Chapter 5

Conclusion

This chapter summarises the principal findings, states the main limitations, and outlines implications for interpreting voluntary health-screening data.

This report analysed 15,219 voluntary health-screening records from a 2006 Austrian exhibition dataset. The work proceeded from data quality assessment and exploratory analysis (Chapter 2) through Random Forest modelling (Chapter 3) to regional and neighbourhood-level spatial analysis (Chapter 4).

5.1 Principal Findings

Three conclusions emerge across the analysis.

Data quality is adequate but not unproblematic. Complete-case cleaning removed 7.1% of records; remaining issues include implausible entries, joint missingness in core fields, and heavy geographic concentration in Styria. These characteristics constrain generalisation beyond the screened population.

Age dominates blood pressure variation; prediction from available covariates remains limited. Exploratory comparisons and Random Forest models consistently identify age as the strongest predictor. Male sex, known blood sugar, and under treatment are associated with higher blood pressure; self-reported wellbeing and several lifestyle flags contribute little after adjustment. Models capture broad population trends—particularly the age–blood pressure gradient—but explain at most 19% of systolic variance, with mean errors of 10–14 mmHg. Hypertension classification offers a complementary risk-category view but suffers from low precision. The available feature set is therefore informative for describing group differences, not for reliable individual-level forecasting.

Geographic context adds limited explanatory value. Regional GDP does not significantly predict state-level mean blood pressure, although it correlates strongly with lower overweight prevalence. Neighbourhood mean systolic pressure enters contextual models with a positive coefficient, yet only 1.8% of variance lies between postal codes. Spatial findings supplement the individual-level analysis but do not alter its central message.

5.2 Limitations

Several constraints should be borne in mind when interpreting these results. The data are cross-sectional and voluntarily collected at exhibition terminals, which limits causal inference and national representativeness. Treatment status may reflect existing hypertension rather than

causing elevated readings. Regional analyses aggregate to nine states with corresponding ecological fallacy risk. Terminal-specific measurement effects were not examined despite known equipment changes during the study period.

5.3 Closing Remarks

The project objectives—characterising data quality, identifying blood pressure determinants, and extending the analysis spatially—were addressed, but with sobering predictive limits. The most reliable insight is descriptive: blood pressure in this sample rises markedly with age, while self-reported risk flags and regional indicators add comparatively little explanatory or predictive power. Future work on this dataset would benefit from terminal-level validation, richer clinical covariates, and prospective designs that support stronger causal claims.

Bibliography

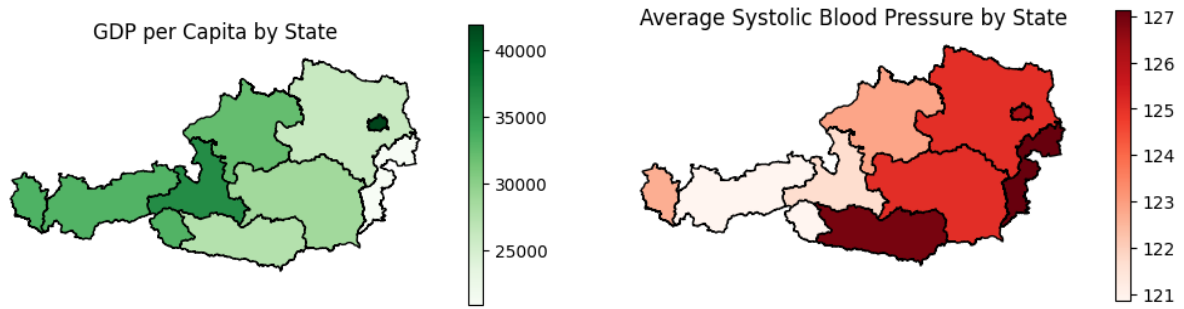
- [1] Course Materials. Styrian health data case study, 2006. Voluntary health-screening dataset from public exhibition terminals.
- [2] European Commission, Directorate-General for Health and Consumer Protection. Cardiovascular risk factors and health monitoring in austria: Final project report, annex. https://ec.europa.eu/health/ph_projects/2002/monitoring/fp_monitoring_2002_annexe_aust_04_en.pdf, 2004. State-level body mass index and overweight prevalence indicators.
- [3] igismap. Download austria shapefile – administrative boundary polygon, rail, road line, ferry, airport point. <https://www.igismap.com/download-austria-shapefile-administrative-boundary-polygon-rail-road-line-ferry-airport-point/>, 2023. Administrative boundary shapefiles for choropleth maps.
- [4] Open-Meteo. Historical weather api. <https://open-meteo.com/>. Hourly temperature and humidity for Graz, Austria, 2006.
- [5] Statistics Austria. Regional atlas. <https://www.statistik.at/atlas/>. GDP per capita and disposable household income at NUTS level 2.
- [6] C. Temml, A. Rieder, A. Thomas, and B. Brüggengjürgen. Development of blood pressure values of an austrian population between 2007–2018: results of comprehensive nationwide examinations. *Wiener Klinische Wochenschrift*, 133:1123–1132, 2021.

Appendix A

Supplementary Materials

This appendix presents supplementary spatial figures referenced in Chapter 4. Choropleth and scatter plots use Austrian administrative boundaries from [3] and regional socioeconomic indicators from [5, 2].

A.1 Supplementary Figures: Socioeconomic Factors



(a) Regional GDP per capita by Austrian state.

(b) Average systolic blood pressure by state.

Figure A.1: Comparison of GDP per capita (left) and average systolic blood pressure (right) across Austrian states based on the health screening dataset. These plots illustrate potential regional variation in socioeconomic status and cardiovascular health.

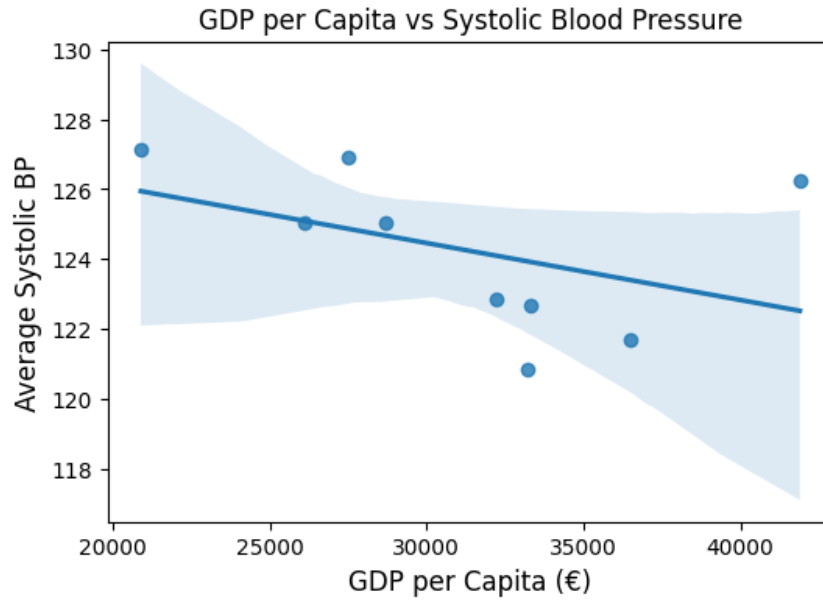


Figure A.2: Scatter plot of regional GDP per capita versus mean systolic blood pressure by Austrian state. Each point represents one state from the health screening dataset, illustrating the absence of a strong regional association between economic prosperity and average systolic blood pressure.

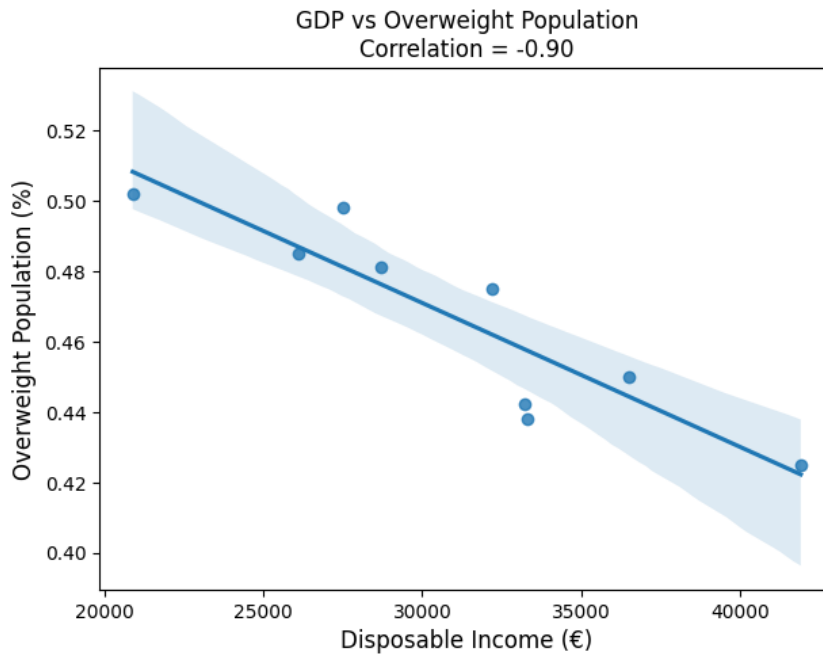


Figure A.3: Scatter plot of regional GDP per capita versus overweight prevalence (proportion of population with $\text{BMI} \geq 25$) by Austrian state. Each point represents one state, showing the strong negative association between economic prosperity and population-level overweight prevalence observed in the health screening dataset and public health statistics.

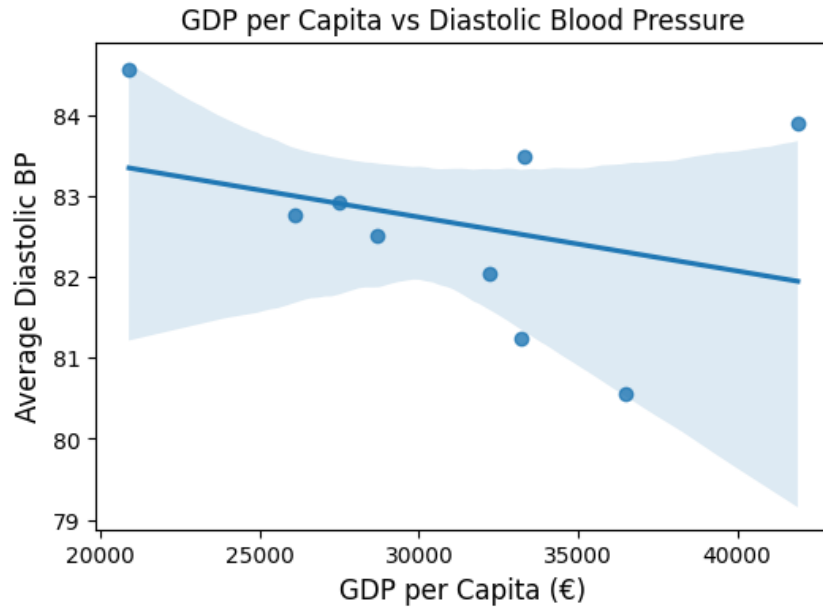


Figure A.4: Scatter plot of regional GDP per capita versus mean diastolic blood pressure by Austrian state. Each point represents one state, illustrating whether regional economic prosperity is associated with variation in average diastolic blood pressure across Austria.

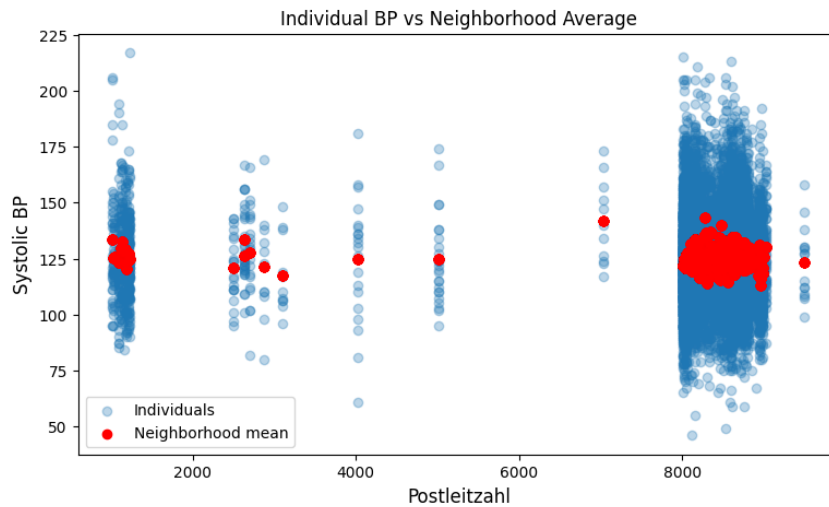


Figure A.5: Scatter plot of neighbourhood mean systolic blood pressure versus individual diastolic blood pressure. Each point represents an individual's measurement, coloured or grouped by neighbourhood (postal code). This figure visualises the contextual relationship explored in the NEAP regression analysis, supporting the modest but non-negligible association between neighbourhood systolic averages and individual diastolic outcomes.

Appendix B

AI Usage Declaration

I declare that I used ChatGPT (OpenAI GPT-4) to assist with structuring the report, improving clarity of explanations, and polishing the academic writing of several sections. In addition it was used to refactor, add inline documentation and to improve the readability of the code.

In addition cursor agents were used to suggest improvements in section organization, to help draft interpretations of figures and tables based on results I had already generated, refining L^AT_EX formatting, and to assist in wording the discussion and future work sections.

The design of the research question, data preprocessing, feature engineering, implementation of the code and models, experimental setup, statistical evaluation are entirely my own work. All final decisions regarding methodology, analysis, and conclusions were made independently by me.